Ryan Dang

(631)-662-1148 || ryan_dang@brown.edu

EDUCATION

Brown University, *B.S. Computer Engineering, GPA 4.0* Expected Graduation May 2027

- Relevant Courses: Digital Design, Computer Architecture, VLSI, Signals and Systems, Dynamics and Vibrations, Calc II & III, Object Oriented Design, Data Structures and Algorithms, E&M, Linear Algebra, Circuits, Embedded Systems

EXPERIENCE

Wright State University Research Experience for Undergraduates (REU) June 2025 - August 2025

Hardware Security Research Intern Dayton, OH

- Implemented a Ring Oscillator Network (RON) on a **Basys 3** Artix-7 FPGA in **Vivado** for hardware Trojan detection using side-channel analysis and measuring frequencies to detect localized power anomalies
- Applied machine learning (SVM and Random Forest) with **scikit-learn** and **pandas** for hardware Trojan detection, achieving 98.8% accuracy and 2% false positive rate, improving robustness over prior methods

Brown University Engineering Department January 2025 - May 2025

Dynamics and Vibrations Teaching Assistant Providence, RI

- Led office hours for **20+** students to clarify concepts in kinematics, force balance, and harmonic motion

AA Technology May 2024 - July 2024

PCB Manufacturing Engineer Intern Ronkonkoma, NY

- Increased conformal coating throughput by **1.5x** by optimizing motion paths and programming for a robotic arm
- Achieved a **5%** reduction in lead time by interpreting PCB schematics, developing a bill of materials in **Excel,** and cross-referencing customer part specifications with datasheets to ensure accuracy with design requirements

PROJECTS

FPGA Pong Game November 2025 - December 2025

- Implemented Pong on a **Cyclone V** FPGA, using **Verilog** to create modular components for paddle control, ball movement, VGA display synchronization, and game state tracking

Remote Door Lock November 2025 - December 2025

- Built lock on Arduino Uno R4 WiFi by hosting web server that receives HTTPS POST requests to control a servo

DMalloc Custom Memory Allocator March 2025 - April 2025

- Implemented a custom **C++** debugging allocator in **Docker** with heap metadata, overflow-safe allocation, boundary write detection, double-free checks, leak reporting, and performance optimizations supporting **500K+** allocations

Tetris and Custom Controller January 2024 - February 2024

- Developed a Tetris clone from scratch in **Java** with custom game logic, collision detection, scoreboard, and a physical controller using a custom PCB (**Altium**) and 3D-printed housing (**Fusion 360**)

ACTIVITIES

Engineering Department Undergraduate Group Providence, RI

Events Coordinator October 2024 - Present

- Increased Engineering Week engagement by **30%** through interactive events, such as trivia night and Lego challenges

Brown Formula Racing Providence, RI

Electrical Engineer January 2024 - Present

- Reduced wiring complexity and volume by **10%** by redesigning the power distribution system via a custom PCB
- Placed **top 20** in the US, assembled a custom wiring harness for the car using DTM and ring connectors

SKILLS

Technical: FPGA, PCB Design, Arduino, x86, TCP/IP, UDP, I2C, SPI, UART, RA4M1, ESP32
Languages/Tools: Verilog, C/C++, Python, Go, Java, Vivado, Quartus, Altium, Cadence, LTSpice, Linux, Docker, Git
Instruments: Oscilloscope, Signal Analyzer, Waveform Generator, Multimeter
Interests: Brown Men's Volleyball, Piano, Guitar, Biking